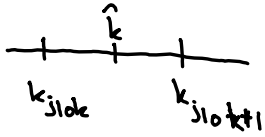


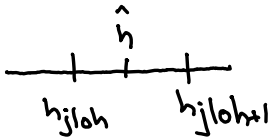
03-17-2025

Two-Dimensional Linear Interpolation:

↳ $V(\hat{k}, \hat{h})$: locate on grid:



$$w_k = \frac{\hat{k} - k_{jlok}}{k_{jlok+1} - k_{jlok}}$$



$$w_h = \frac{\hat{h} - h_{jloh}}{h_{jloh+1} - h_{jloh}}$$

① fix h_{jloh}

$$[(1-w_k)V(jlok, jloh) + w_k V(jlok+1, jloh)] (1-w_h) +$$

② fix h_{jloh+1}

$$[(1-w_k)V(jlok, jloh+1) + w_k V(jlok+1, jloh+1)] \cdot w_h$$

] stochastic variable z :

↳ discretize: $E v(k, z) = \sum_{i=1}^n p_i v(k, z_i)$

i.i.d $z \sim N(\mu, \sigma_z^2) \Rightarrow z \in [z_1, \dots, z_n]$ w/ $p_1, \dots, p_n$

1. Set n

2. Set $\bar{z} = \mu + \lambda \cdot \sigma_z$; $\underline{z} = \mu - \lambda \sigma_z$

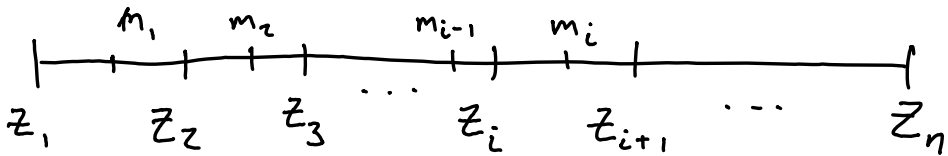
3. Set $z_2, \dots, z_{n-1}$

3. Set $z_2, \dots, z_{n-1}$

$$z_1 = \underline{z}; z_2 = \underline{z} + \text{step}; z_3 = \underline{z} + 2\text{step}; \dots; z_n = \overline{z}$$

$$\text{Step} = \frac{\overline{z} - \underline{z}}{n-1} = \frac{2\lambda\sigma_z}{n-1}$$

4.



$$m_i = \frac{z_i + z_{i+1}}{2}$$

$$\begin{aligned} \Pr(z = z_i) &= \Pr(z \in [m_{i-1}, m_i]) - \Pr(m_{i-1} < z < m_i) \\ &= \Pr\left(\frac{m_{i-1} - \mu}{\sigma_z} < \frac{z - \mu}{\sigma_z} < \frac{m_i - \mu}{\sigma_z}\right) \\ &= \Pr\left(\frac{z - \mu}{\sigma_z} < \frac{m_i - \mu}{\sigma_z}\right) - \Pr\left(\frac{z - \mu}{\sigma_z} < \frac{m_{i-1} - \mu}{\sigma_z}\right) \\ \text{CDF} \swarrow &= \Phi\left(\frac{m_i - \mu}{\sigma_z}\right) - \Phi\left(\frac{m_{i-1} - \mu}{\sigma_z}\right) \end{aligned}$$

$$\Pr(z = z_1) = \Pr(z < m_1) = \Phi\left(\frac{m_1 - \mu}{\sigma_z}\right)$$

$$\Pr(z = z_n) = \Pr(z > m_{n-1}) = 1 - \Phi\left(\frac{m_{n-1} - \mu}{\sigma_z}\right)$$

Tauchen (1986)

$$\hookrightarrow z_{t+1} = \mu(1-\rho) + \rho z_t + \varepsilon_{t+1} \Rightarrow \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2)$$

$$\Pr(z_{t+1} = z_i | z_t = z_j)$$

$\hookrightarrow$ Conditional Dist. of z :

$$\left. \begin{aligned} E[z_{t+1} | z_t] &= \mu(1-\rho) + \rho z_t \\ \text{var}(z_{t+1} | z_t) &= \sigma_\varepsilon^2 \end{aligned} \right\} z_{t+1} | z_t \sim N(\mu(1-\rho) + \rho z_t, \sigma_\varepsilon^2)$$

$\hookrightarrow$ Unconditional Dist of z :

$$\left. \begin{aligned} E[z] &= \mu \\ \text{var}(z) &= \frac{\sigma_\varepsilon^2}{1-\rho^2} \end{aligned} \right\} z \sim N\left(\mu, \frac{\sigma_\varepsilon^2}{1-\rho^2}\right)$$

1. Set n

2. Set $\underline{z} = \mu - \lambda \sigma_z$; $\bar{z} = \mu + \lambda \sigma_z$

3. Set $z_2, \dots, z_{n-1} = z_i + \text{step}$ s.t. $\text{step} = \frac{2\lambda \sigma_z}{n-1}$

4. Construct midpoints

$$\text{Ex: } \Pr(z' = z_i | z = z_s)$$

$$= \Pr\left(\frac{m_{i-1} - (1-\rho)\mu - \rho z_s}{\sigma_\varepsilon} < \frac{z' - (1-\rho)\mu - \rho z_s}{\sigma_\varepsilon} < \frac{m_i - (1-\rho)\mu - \rho z_s}{\sigma_\varepsilon}\right)$$

$$= \Pr\left(\frac{z' - (1-\rho)\mu - \rho z_s}{\sigma_\varepsilon} < \frac{m_i - (1-\rho)\mu - \rho z_s}{\sigma_\varepsilon}\right) - \Pr\left(\frac{z' - (1-\rho)\mu - \rho z_s}{\sigma_\varepsilon} < \frac{m_{i-1} - (1-\rho)\mu - \rho z_s}{\sigma_\varepsilon}\right)$$

$$= \Phi\left(\frac{m_i - (1-\rho)\mu - \rho z_s}{\sigma_\varepsilon}\right) - \Phi\left(\frac{m_{i-1} - (1-\rho)\mu - \rho z_s}{\sigma_\varepsilon}\right)$$

$$Pr(z=z_1 | z_s) = \Phi \left(\frac{m_1 - (1-p)\mu - pz_s}{\sigma_\varepsilon} \right)$$

$$Pr(z=z_n | z_s) = 1 - \Phi \left(\frac{m_{n-1} - (1-p)\mu - pz_s}{\sigma_\varepsilon} \right)$$

Tauchen-Hussey (1991)

↳ Gauss-Hermite quadrature

$$\int_{-\infty}^{\infty} g(x) e^{-x^2} dx \approx \sum_{i=1}^n w_i g(x_i)$$

$$\int v(z') f(z'|z) dz' = \int v(z') \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} e^{-\frac{(z' - \mu(1-\rho) - \rho z)^2}{2\sigma_\varepsilon^2}} dz'$$
$$\Rightarrow \underbrace{v(z') f(z'|z)}_{\frac{f(z'|\alpha)}{f(z'|\mu)}} = \varphi(z', z) \cdot f(z'|\alpha)$$

↳ $\alpha = \mu$

$$\Rightarrow f(z'|z=\mu) = \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} e^{-\frac{z'-\mu}{2\sigma_\varepsilon^2}}$$

$$\hookrightarrow z'|z=\mu = N(\mu(1-\rho) + \rho\mu, \sigma_\varepsilon^2) = N(\mu, \sigma_\varepsilon^2)$$

$$\hookrightarrow y = \frac{z-\mu}{\sqrt{2\sigma_\varepsilon^2}}; z = \sqrt{2\sigma_\varepsilon^2} y + \mu; dz = \sqrt{2\sigma_\varepsilon^2} dy$$

$$\int \varphi(z', z) f(z'|\mu) dz' = \int \varphi(z', z) \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} e^{-\frac{z'-\mu}{2\sigma_\varepsilon^2}} dz'$$
$$= \int \varphi(\sqrt{2\sigma_\varepsilon^2} y + \mu, z) \sqrt{2\pi\sigma_\varepsilon^2} e^{-y^2} \sqrt{2\sigma_\varepsilon^2} dy$$
$$= \frac{1}{\sqrt{\pi}} \int \varphi(\sqrt{2\sigma_\varepsilon^2} y + \mu, z) e^{-y^2} dy$$

$$= \frac{1}{\sqrt{\pi}} \int \varphi(\sqrt{z} \sigma_{\varepsilon} y + \mu, z) e^{-y^2} dy$$

$$= \frac{1}{\sqrt{\pi}} \sum_{i=1}^n \varphi(\sqrt{z} \sigma_{\varepsilon} y_i + \mu, z_t) \cdot \omega_i$$

$$= \frac{1}{\sqrt{\pi}} \sum_{i=1}^n v(\sqrt{z} \sigma_{\varepsilon} y_i + \mu) \cdot \frac{f(z_i | z_t)}{f(z_i | \mu)} \cdot \omega_i$$

$$\tilde{p}(z=z_i | z_t) = \frac{1}{\sqrt{\pi}} \cdot \omega_i \cdot \frac{f(z_i | z_t)}{f(z_i | \mu)}$$

$$p(z=z_i | z_t) = \tilde{p}(z=z_i | z_t) / \sum_{i=1}^n \tilde{p}(z=z_i | z_t)$$

$$\int v_i(z') f(z' | z) \approx \sum_{i=1}^n p(z=z_i | z_t) v(z_i)$$